Detecting AI-Generated Responses in Surveys

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Abstract

Generative AI poses a measurement problem for survey research: when respondents can outsource summarization, explanation, or recall to a chatbot, open-ended responses may reflect tool access as much as the respondent's own thinking. We introduce an auditable toolkit that detects likely AI-assisted completion by combining response-process paradata with prompt-specific semantic benchmarks. The toolkit integrates keystroke and timing measures, deletion and revision traces, and embedding-based text diagnostics, including similarity to an AI benchmark response and peer-distinctiveness measures. We validate the approach in two ways. First, we field a randomized survey experiment with article-summary tasks in which respondents are assigned either to a blocked-AI control condition or to AI-access conditions that vary in assistance quality. Second, we benchmark the same workflow against end-to-end completions generated by an AI agent. Across both validations, AI-assisted or delegated responses show thinner writing traces, higher benchmark similarity, and lower peer originality. The resulting classifiers achieve strong out-of-sample performance across conditions.

Keywords: survey methodology; AI; survey authenticity; paradata; classification

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1 Introduction

The rapid spread of large language models has created a new authorship and measurement problem for survey research. Respondents can now use AI to draft, polish, or replace their answers, and autonomous agents can complete entire surveys with little or no human involvement. These are related but not identical problems. A fully automated response raises a problem of deception: there may be no respondent behind the record at all. AI-assisted responding raises a problem of delegation: a real respondent is present, but part of the cognitive work the survey is meant to elicit has been outsourced. For survey methodology, both matter because they can produce answers that appear thoughtful and on-topic while reflecting very different response processes.

This is especially consequential for open-ended questions. Researchers often use these items not simply to collect text, but to infer that respondents processed a stimulus, remembered information, articulated considerations in their own words, or reasoned through a task. Once AI enters that process, surface fluency is no longer a reliable proxy for those underlying behaviors. The scale of the problem is already substantial: roughly a third of crowdworkers report using LLMs on open-ended items (Zhang, Xu, and Alvero 2025), autonomous AI agents pass standard survey quality checks at high rates (Westwood, n.d.), and a recent audit of a live online panel flagged at least 4.4% of respondents as likely AI-assisted using conservative criteria (Westwood and Frederick, n.d.). Survey researchers should therefore expect that open-ended data increasingly combine unaided human writing, human-AI collaboration, and fully automated generation.

Existing safeguards were not built for this mixture. Attention checks, CAPTCHAs, and speed flags were designed to catch careless humans and crude bots, not respondents who strategically outsource their thinking to a language model (Park et al. 2025; Höhne et al. 2025; Westwood, n.d.). Off-the-shelf AI-text detectors are unreliable, degrade as models improve, and cannot tell researchers whether AI use invalidates a response for the specific construct a question is meant to capture (Crossley et al. 2024; Yan et al.,

n.d.). Specialized classifiers can perform well in the settings on which they were trained, but often lose accuracy across new questions, topics, and prompt formats (Claassen et al. 2026). The problem, then, is not only one of fraud detection. It is also a problem of identifying when different forms of authorship leave different traces in survey data, and when those traces matter for inference.

This paper presents evidence that human, human-AI, and AI-generated responses leave systematically different traces in surveys, and it develops a toolkit researchers can use to study, flag, and report those differences. Open-ended items are methodologically useful here because they preserve evidence not only in the final answer but also in the process by which that answer was produced. An instrumented text box can record paste attempts, keystrokes, deletions, pauses, and timing, while the text itself can be compared to the source material, to peer responses, and to prompt-specific AI counterfactuals. By combining process traces with textual diagnostics, researchers can recover meaningful signals of authorship inside an ordinary survey. The novelty lies in integrating front-end instrumentation, interpretable writing-process measures, prompt-specific semantic benchmarks, and model-based classification into a single workflow that can be deployed within an ordinary survey. Building on recent calls for layered defenses against AI survey contamination (Panizza, Kyrychenko, and Roozenbeek 2026), we develop an auditable process-plus-product toolkit that can be adapted to future studies.

We frame that toolkit as a measurement aid rather than a universal exclusion rule. Whether AI assistance makes a response unusable depends on the construct of interest. For some tasks, the final judgment may matter most. For others, researchers care about recall, comprehension, reasoning, or spontaneous articulation in a respondent's own words. Our contribution is therefore not to claim that any AI involvement is automatically invalid, nor to offer a detector that should travel unchanged across all survey contexts. Instead, we provide a flexible workflow that helps researchers assess when an observed response no longer matches the response process their design assumes, flag likely cases transparently, and identify which prompts are especially prone to delegation. In

that sense, the toolkit is useful not only for data cleaning but also for survey redesign. Furthermore, we encourage researchers to use AI agents as a prefielding stress test and provide a template for incorporating them into survey design evaluation.

We validate that workflow in two ways. First, we use a randomized experiment that varies AI access on an article-summary task. This improves on observational audits and proxy-label designs because it creates a known difference in response-production conditions while preserving a task that remains substantively meaningful for survey research. Second, we benchmark the same instrument against responses produced by a synthetic AI agent that completes the survey end-to-end, allowing direct comparison between human AI assistance and fully automated completion on a common set of behavioral and semantic diagnostics. Across these settings, AI use changes both how answers are produced and what they look like: assisted respondents and autonomous agents show less visible drafting inside the response box, produce fewer signs of revision, and generate text that is closer to prompt-specific AI benchmarks and less semantically distinctive from peer responses.

The paper, therefore, contributes both evidence and tools. Empirically, it shows that human, human-AI, and AI-generated survey responses leave measurably different behavioral and textual traces. Methodologically, it offers a toolkit researchers can adapt to their own instruments, combining front-end instrumentation, interpretable writing-process indicators, prompt-specific semantic benchmarks, evaluator scores, creativity and originality diagnostics, summary embeddings, and classification models. We also provide a machine-learning classifier based on our toolkit and show how performance varies across these feature bundles. The paper also contributes a practical template for using AI agents before fielding as a stress test for survey instruments, helping researchers identify prompts that are especially easy to delegate and diagnose where design changes may be needed. Our broader recommendation is that researchers treat authorship as part of measurement, report AI-flagging procedures and rates transparently, and use these tools both to evaluate data quality and to redesign tasks or response modes when delegation

becomes easier than genuine participation.

2 Motivation

Recent discussions in survey methodology and political methodology converge on a common point: credible online data collection now depends less on any single screening device than on transparent workflows that document how suspicious cases are identified and handled. Reviews of survey fraud show that the usable yield of low-barrier web recruitment has deteriorated sharply and that standard defensive batteries are readily bypassed (Pinzón et al. 2024). Work in political methodology similarly shows that explicit combinations of automated and human-coded checks can recover much of the data quality achieved by intensive supervision (Cohen and Warner 2021), while studies of targeted populations document how adversarial respondents learn to defeat simple screeners (Bell and Gift 2023). The lesson is not simply that online data are noisy. It is that valid inference increasingly depends on observable, reviewable quality-control procedures.

AI assistance sharpens that requirement because it introduces a different kind of measurement failure. One useful distinction is between deception and delegation: sometimes no real respondent stands behind the record, and sometimes a real respondent is present but outsources part of the task to AI. These cases call for overlapping but not identical responses. Classic fraud and satisficing often degrade answers in visible ways, such as speeding, straightlining, contradiction, or nonsense. AI-assisted delegation can do the opposite. It can improve the visible product while weakening the link between the response and the respondent's own reasoning. That matters for the open-ended tasks political scientists use to evaluate learning, comprehension, attribution, or justification. The measurement problem is therefore not only whether a respondent is human, but whether the observed answer was produced under conditions consistent with the construct being measured (Zhang, Xu, and Alvero 2025; Westwood, n.d.; Westwood and Frederick, n.d.). Not every instance of AI use is equally damaging, but when a task is meant to capture

unaided cognition, authorship becomes part of measurement.

This paper also differs from an adjacent literature that asks when LLMs can validly stand in for human subjects. That work focuses on calibration, substitution, and the inferential conditions under which simulated responses can be useful for exploratory or confirmatory research (Hullman et al. 2026). Our question is different: how should researchers study and handle AI assistance in surveys designed to measure human responses? That difference motivates a broader approach than a generic text detector or a one-off threshold, because contamination can appear as direct copying, partial borrowing, scaffolded drafting, or heavily edited transcription. We therefore combine preventive design, process paradata, and task-specific semantic diagnostics in a toolkit that helps researchers both flag likely assistance transparently and identify where the survey itself may be inviting delegation.

3 An Auditable Toolkit for Authenticating Human Survey Completion

The workflow is designed as an auditable alternative to two incomplete strategies in the current literature: text-only detection and narrow behavioral heuristics. Text-only models can classify machine-written narratives in question-specific corpora, but their performance often weakens across prompts and they reveal little about how an answer was produced (Claassen et al. 2026). Simple audits based only on paste events or extreme typing speeds can be useful first passes, but they can miss more selective forms of assistance (Westwood and Frederick, n.d.). Our approach, therefore, organizes evidence into the following indicators: *Behavior*, *AI signature*, *Evaluation*, and *Creativity and originality*. We then extend that core toolkit in two directions. First, we add *Text embedding*, which captures the semantic content of the response itself in a dense representation. Second, in tasks that contain an external source text, we add *Text alignment*, which compares the response to that reference material. The aim is not to generate a universal black-box

verdict. It is to preserve enough information that researchers can explain why a response appears authentic, suspicious, or ambiguous.

At a substantive level, the proposal is straightforward. Open-ended survey questions can serve not only as substantive outcomes, but also as authentication devices, because they expose traces of composition that closed-ended items usually hide. The relevant evidence is multidimensional: how the response was composed, what semantic content it contains, how near it lies to a prompt-specific AI counterfactual, how distinctive it remains relative to peers, whether evaluator scores imply unusual quality gains, and, where applicable, whether the response remains anchored to a common source text. Table 1 summarizes the measure families and their role in the workflow.

Table 1: Toolkit measure families

Measure family	Illustrative measures	Authentication role
Behavior	Paste indicator, paste count, task time, seconds per word, keystrokes, extra keys, deletions, key-to-character ratio	Captures whether the response appears to have been drafted inside the survey environment rather than transferred in or produced through unusually linear composition.
Text embedding	Summary-text embedding dimensions, sentence embedding representation of the response	Encodes the response itself as a dense semantic representation, allowing the classifier to detect broader semantic regularities not captured by targeted hand-crafted measures.
AI signature	Similarity to benchmark AI answer, title similarity to benchmark AI answer, novelty relative to AI benchmark	Measures proximity to a prompt-specific machine counterfactual rather than to an abstract notion of "AI writing."
Evaluation	Quality index, clarity/coherence, accuracy, typo indicators	Helps interpret whether quality gains are accompanied by other signs of outsourcing; used as supplementary evidence rather than a standalone detector.
Creativity and originality	Peer originality, peer distance, creativity index, lexical diversity, n-gram novelty	Captures whether a response remains distinctive relative to other respondents or collapses toward highly standardized output.
Text alignment	Similarity to text reference (if available)	Evaluates whether the response remains anchored to assigned source material. This category is task-specific and is only applicable when respondents are writing about a common reference text.

Behavior. The first category captures the production process itself. This block includes direct transfer indicators such as paste events, but also broader signs of how a response was composed: task time, seconds per word, keystrokes, extra keys, deletion activity, and key-to-character ratios. The rationale is not that any one of these variables is dispositive. Rather, together they capture whether the respondent appears to have drafted the answer in the survey box through an iterative process of hesitation, revision, and correction, or whether the text entered the box with a much thinner production history. These measures are attractive because they are both interpretable and close to the survey action researchers actually care about: whether the respondent composed the answer in the survey environment.

Text embedding. The second category represents the response itself as a high-dimensional semantic object. This category embeds the response text and lets a classifier model learn broader patterns in semantic content and style that may be difficult to summarize with a few hand-crafted measures. In practical terms, this category is useful because it preserves information that targeted variables may miss. A response may not be extremely close to the AI benchmark, may not show a clear paste signature, and may still occupy a part of the semantic space that is systematically associated with assisted completion. The tradeoff is that embedding dimensions are less directly interpretable one by one. Their value lies in preserving richer semantic structure, not in providing a single transparent diagnostic on their own.

Evaluation. The *Evaluation* category includes clarity/coherence, accuracy, typo indicators, and a composite quality index. These measures are useful, but they play a different role from the core behavioral and semantic variables. High-quality writing is not itself evidence of inauthenticity, and lower-quality writing is not evidence of authenticity. Instead, evaluator scores help interpret cases in which semantic similarity or behavioral irregularities coincide with unusually large quality gains. In that sense, evaluator measures are best understood as supplementary evidence: they can clarify whether a suspiciously efficient

response also looks exceptionally polished or unusually detached from the respondent's own phrasing, but they should not be treated as standalone detectors. We used an LLM as a judge, relying on the OpenAI API for the evaluation.

AI signature. The *AI signature* category captures proximity to a prompt-specific machine counterfactual. For each prompt, the workflow generates a generic AI answer under the same instruction set and embeds that answer in the same semantic space as the respondent's response (see Figure 1). Measures such as similarity to the benchmark AI answer, title similarity to the benchmark, then quantify how closely the respondent's text converges toward that prompt-specific AI reference. This matters because a generic detector cannot distinguish between answers that look "machine-like" because they were AI assisted and answers that look similar simply because the prompt tightly constrains the space of reasonable summaries. By grounding the comparison in the exact article and instruction set, the benchmark becomes auditable and task-specific rather than universal and opaque.

Creativity and originality. The *Creativity and originality* category captures whether a response remains distinctive relative to the local distribution of peer answers. This block includes both semantic and surface-form measures. On the semantic side, *peer originality* compares a respondent's embedding to the leave-one-out centroid of all other responses to the same prompt and defines originality as one minus cosine similarity, so that larger values indicate greater distinctiveness from the peer center. On the language side, lexical diversity measures such as type-token ratios and Yule's K summarize how repetitive or varied the vocabulary is, while novelty measures based on uncommon n-grams capture whether the respondent uses phrasing that is less formulaic relative to the sample. The rationale is that assistance does not always appear as verbatim copying or obvious transfer. A respondent may partially borrow, scaffold, or lightly edit external text in ways that compress the distribution of responses within a prompt without producing a clear paste signature. These measures therefore ask whether the answer retains distinctive

semantic content and varied language, or whether it collapses toward a more standardized response pattern.

The semantic logic behind the two most distinctive embedding-based measures in the toolkit is illustrated in Figure 1, which places *AI signature* and *peer originality* in a common framework. Both begin by mapping a respondent's answer into the same embedding space. AI signature then compares that embedding to a generic AI answer generated under the same prompt, yielding a prompt-specific measure of AI proximity. Peer originality instead compares the respondent's embedding to the leave-one-out centroid of peer responses, producing a measure of how unusual that answer is relative to the local response distribution. Taken together, the two measures distinguish closeness to a machine counterfactual from distinctiveness relative to peers.

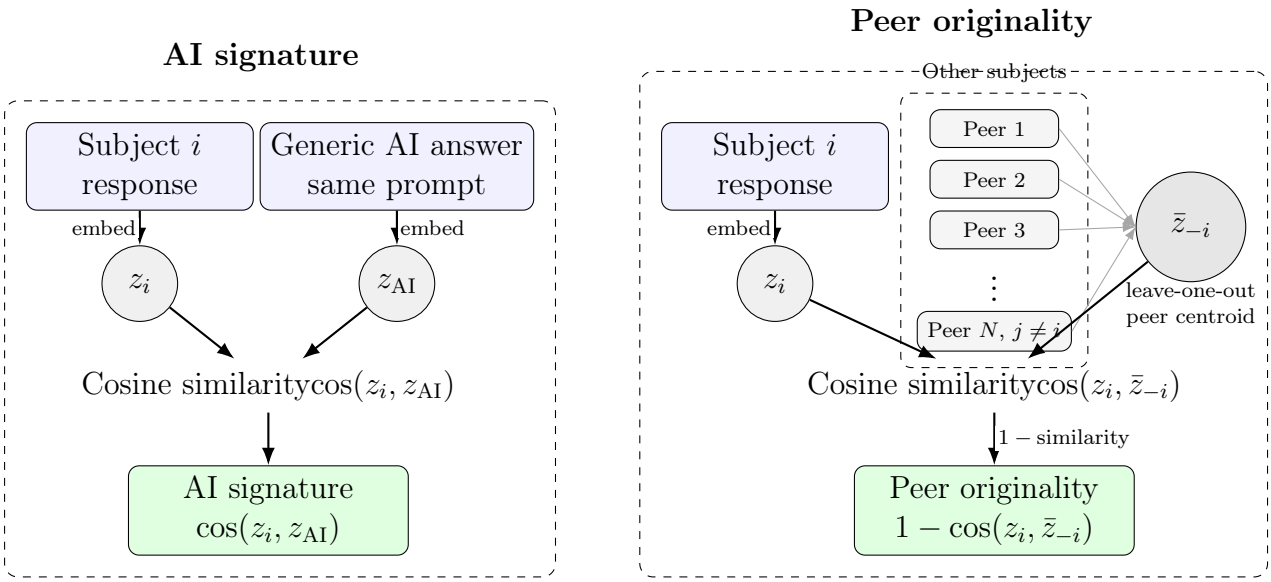

Figure 1: Construction of prompt-specific semantic measures for AI proximity and respondent distinctiveness.

Notes: Both measures begin by mapping a subject's response into a common embedding space. AI signature compares that embedding to the embedding of a generic AI answer generated under the same prompt. Peer originality compares the same subject embedding to the leave-one-out centroid of all other respondents' embeddings and is defined as one minus cosine similarity, so that larger values indicate greater distinctiveness relative to peers.

Text alignment. The final category, *Text alignment*, is intentionally task-specific. In the article-summary setting studied here, respondents are asked to produce a summary

of a common source text, so it is substantively meaningful to compare their answers back to that article. Measures such as similarity to article and faithfulness to article therefore ask whether the response remains anchored to the assigned material rather than drifting toward generic or externally supplied content. This category is useful precisely because it is not universal. In many open-ended survey tasks there is no common reference text, and in those settings text alignment is not applicable by design. The point is not to force every open-ended item into the same template, but to add task-specific diagnostics when the structure of the prompt makes them available.

Taken together, these categories yield an auditable profile rather than a stylometric black box. A respondent may use AI and then retype or lightly edit the answer, muting direct transfer evidence but leaving unusually linear behavioral traces and high benchmark similarity. Another respondent may produce a polished and accurate answer whose evaluator scores are strong, yet whose broader pattern of behavior, embedding position, and AI proximity suggests outsourced completion. In tasks with a common source text, responses may also diverge from the assigned material even when they remain fluent and superficially plausible. The value of the toolkit is therefore cumulative. It does not rest on any single signal. It rests on combining interpretable categories of evidence that map directly onto how open-ended survey responses are actually produced.

4 Research Design and Validation Logic

The research design has two parts. First, we run a randomized survey experiment in which respondents complete an open-ended summarization task under different AI-access conditions. Some respondents are assigned to a blocked-AI control condition and explicitly told that we will screen AI use, while others are assigned to AI-assistance conditions that vary the type and quality of assistance available. We created our own LLM app embedded in the survey, built with Streamlit and based on the OpenAI API.[1] This ex-

1. We track app usage and can link it to respondents.

perimental structure lets us test whether providing AI changes the response process and the final text, rather than inferring AI use from observational proxies alone. Second, we build our own synthetic AI agent and have it complete the same survey instrument end-to-end. That benchmark lets us compare ordinary human AI assistance with a more extreme form of automation and ask whether the same indicators recover both kinds of contamination.

The current human validation sample contains 1,647 respondents recruited through CloudResearch Connect. Respondents completed article-summary tasks and were randomly assigned to one of seven article topics. They saw only one topic. Respondents were also randomized into a blocked-AI control condition ($N = 412$) and three AI-access conditions ($N = 1{,}235$ combined). The main analysis presents AI-only results. Detailed descriptive statistics for the three AI-access conditions are reported in the appendix.

Data collection centers on a substantively meaningful open-ended task rather than an artificial detection exercise. Respondents read a short article, write a summary, and provide a title. This is a useful validation setting because summarization is common in survey research, produces enough text for semantic comparison, and is easy to outsource. The survey records the response process directly through paste events, deletion behavior, keystroke counts, and timing, so the design can capture both the submitted answer and how that answer was produced. Summarization is also substantively conservative: political scientists routinely ask respondents to summarize information, explain what they learned from a treatment, restate a candidate's position, or describe the main point of a news item. A response that looks article-faithful and well written is precisely the kind of answer a researcher might otherwise interpret as evidence of successful comprehension.

The validation design is therefore organized around three related samples rather than a single contrast. First, the stricter `AI_only` subset, which provides a cleaner benchmark for model training and comparison. Second, the pooled randomized human sample identifies the response signature of AI access under ordinary survey conditions. Third, the AI-agent benchmark contributes 543 fully automated completions of the same instrument, allowing

us to compare human AI assistance with end-to-end machine completion on a common set of indicators. When we combine the direct human-assistance benchmark with the AI-agent benchmark, the resulting comparison uses $N = 1{,}371$ observations ($n = 412$ control, $n = 416$ AI-only, $n = 543$ AI agent). This layered structure lets us ask not only whether AI access changes response production, but also whether the same toolkit generalizes from human-assisted responding to fully automated survey completion.

The design is deliberately conservative in what it claims. The primary validation target is experimentally induced AI access, not perfectly observed use in every instance. The AI-agent benchmark then asks whether those same indicators also recover the more extreme case of end-to-end automation. The note, therefore, does not argue that it has solved the universal AI-detection problem. It argues instead that experimentally varied AI access and benchmarked automated completion leave observable response signatures and that open-ended items can be instrumented to recover much of that signal in an auditable way inside broader surveys.

5 Validation Results under Randomized AI Access

Figure 2 presents AI assistance against the blocked-AI control group and reports topic-adjusted standardized contrasts. Across these outcomes, AI access is associated with longer submitted text, substantially more paste activity, fewer keystrokes relative to the amount of text retained, and shorter completion times. The key point is not simply that assisted responses look different at the point of submission. They are produced through a different workflow. Relative to blocked access, the behavioral trace shifts away from iterative drafting inside the survey box and toward transfer, light editing, and efficient transcription of externally generated text.

The semantic plots point in the same direction. Figure 2 shows that AI-exposed responses move closer to the study-specific AI benchmark and that peer distinctiveness also decline. Furthermore, the evaluator identifies fewer typos in AI-assisted answers

relative to the control. This is precisely the pattern that creates a survey-validity problem: the final answer can remain on topic and look competent while becoming less diagnostic of the respondent's own processing.

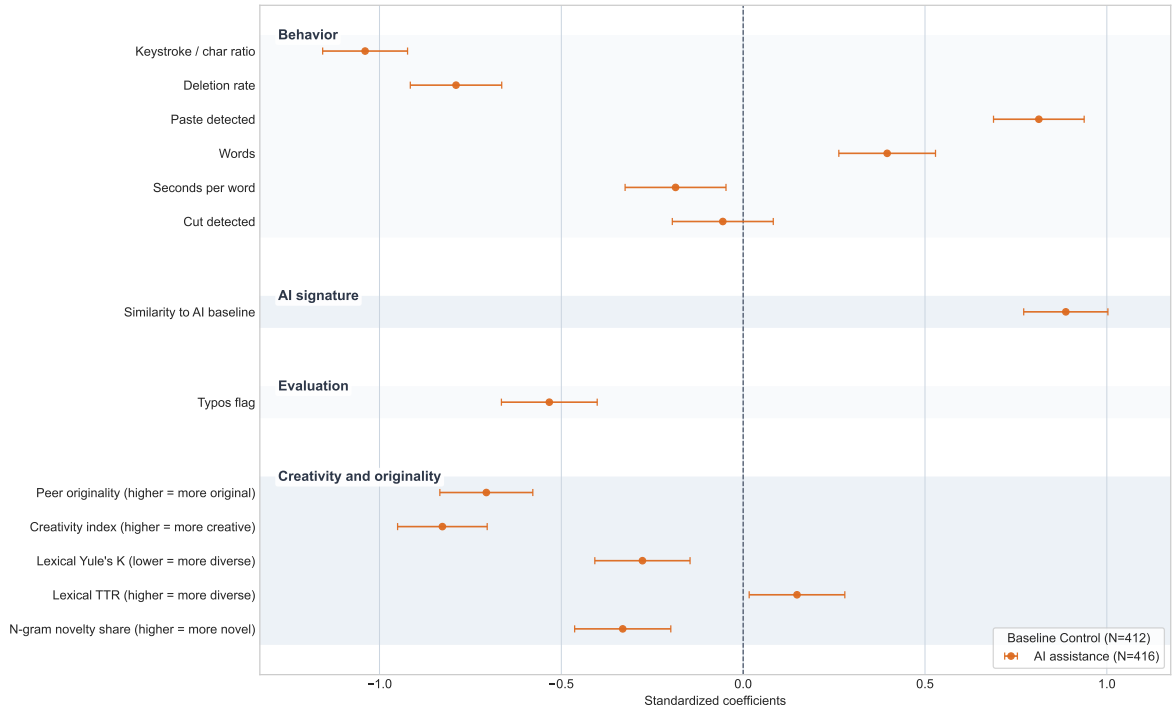

Figure 2: Standardized differences in summary-task outcomes between the human AI-only treatment and the human control group.
Note: Each point is the coefficient on an indicator for assignment to the AI assistance arm from a separate topic-adjusted OLS regression with article-topic fixed effects; horizontal bars show 95% HC1 confidence intervals. All dependent variables are standardized, so coefficients are in standard-deviation units rather than raw units. Positive coefficients indicate higher values in `AI assisted` than in the baseline control group; negative coefficients indicate lower values. The *Behavior* category captures the writing process: *Words* is final summary length, *Seconds per word* captures production speed, *Keystroke / char ratio* and *Deletion rate* capture editing effort, and *Paste detected* and *Cut detected* are binary indicators for whether those actions occurred during writing. The *AI signature* measure is similarity to an AI baseline summary. The *Evaluation* measure, *Typos flag*, is a binary indicator coded one when evaluators flagged typos. The *Creativity and originality* outcomes capture peer-rated originality and creativity, lexical diversity (*Lexical TTR*; *Yule's K*, where lower raw values imply greater diversity), and n-gram novelty. Refer to Figure 9 for additional measures, and refer to Figure 8 for other AI arms such as low-quality AI provided.

AI access often makes responses more homogeneous, even though the language can be more diverse; they are well-drafted, longer, and reduce evidence of respondent effort. If a researcher looked only at final text, AI-exposed respondents could appear more articulate, more accurate, or more engaged. The paradata and novelty suggest a different interpretation: some of the apparent improvement reflects changes in production conditions rather

than deeper processing of the stimulus.

Classifier Performance. Besides providing the tooling, we also train a model to predict AI usage. This model also allows us to predict which of these indicators contributes the most to the detection. In the human-only validation sample, the strongest out-of-sample models reach ROC-AUCs of a random forest with an average accuracy of .815 and .865, and the best-performing bundles are behavioral features and the embeddings of the text.

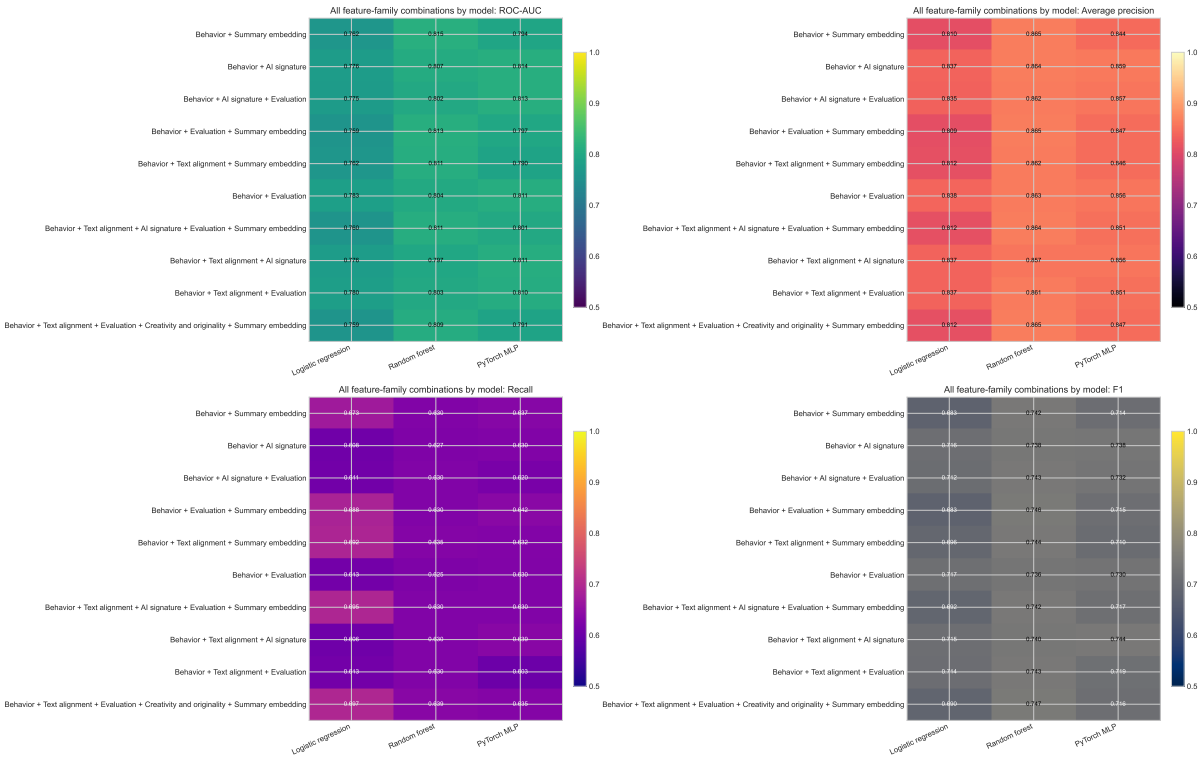

Figure 3: AI-only assistance versus baseline control: top feature-family combinations in classifier performance.

Notes: The figure reports the out-of-sample classifier performance for distinguishing human responses written with AI assistance (`AI_only`) from baseline human control responses. The analysis sample contains $N = 828$ observations ($n = 416$ AI-only, $n = 412$ control). Columns correspond to logistic regression, random forest, and PyTorch MLP models. Rows report only the top 10 feature-family combinations, ordered by the highest ROC-AUC achieved by any model for that combination. The four panels report ROC-AUC, average precision, recall, and F1. Feature families include *Behavior*, *Text alignment*, *AI signature*, *Evaluation*, *Creativity and originality*, and *Summary embedding*. The *Summary embedding* family is based on a 384-dimensional sentence-transformer representation of the written summary (`all-MiniLM-L6-v2`). For the complete set of feature-family combinations, see Appendix Figure 11. For the broader comparison that also includes respondents randomized to low-quality AI assistance, see Appendix Figure 12.

The ranking of single indicators reinforces the same point. The strongest standalone signals are behavioral, followed by AI signature and peer originality. Appendix Table 2 presents the standalone contribution of each indicator.

The appropriate practical use is not to impose a universal exclusion threshold. A better approach is to calibrate study-specific cutoffs on transparent indicators or to generate an AI-likely score from those same measures and then report how the rule was chosen. The workflow is most defensible when the analyst can show, before seeing the substantive results, which indicators matter, why they matter, and how the rule performs in validated data. That approach fits contemporary standards for survey-quality reporting better than reliance on a proprietary detector with no clear decision logic.

6 Validation Results with an AI Agent Benchmark

To extend the validation beyond human respondents who could use AI tools, we administered the same survey to a `SyntheticSurveyAgent` benchmark respondent. The agent was run through the Chrome browser channel and used `gpt-4o-mini` to answer survey items and generate open-text responses. Each synthetic completion was anchored to a pre-specified persona profile so that the benchmark would reflect a plausible survey participant rather than an unconditioned model producing abstract text.

We obtained 543 completed agent responses. For benchmarking, we compare those responses with the same human control group used as the non-AI baseline and with the human AI-assistance group. This puts fully automated completion and assisted human completion on the same scale and shows whether the toolkit is recovering a common response signature rather than overfitting to a single mode of AI use.

Figure 4 shows a consistent ordering across the most diagnostic measures. Human AI assistance already lowers keystroke-to-character ratios, reduces deletion, raises paste incidence, and increases similarity to the AI benchmark relative to unaided human writing. The AI agent pushes those same patterns further: it shows the thinnest drafting trace,

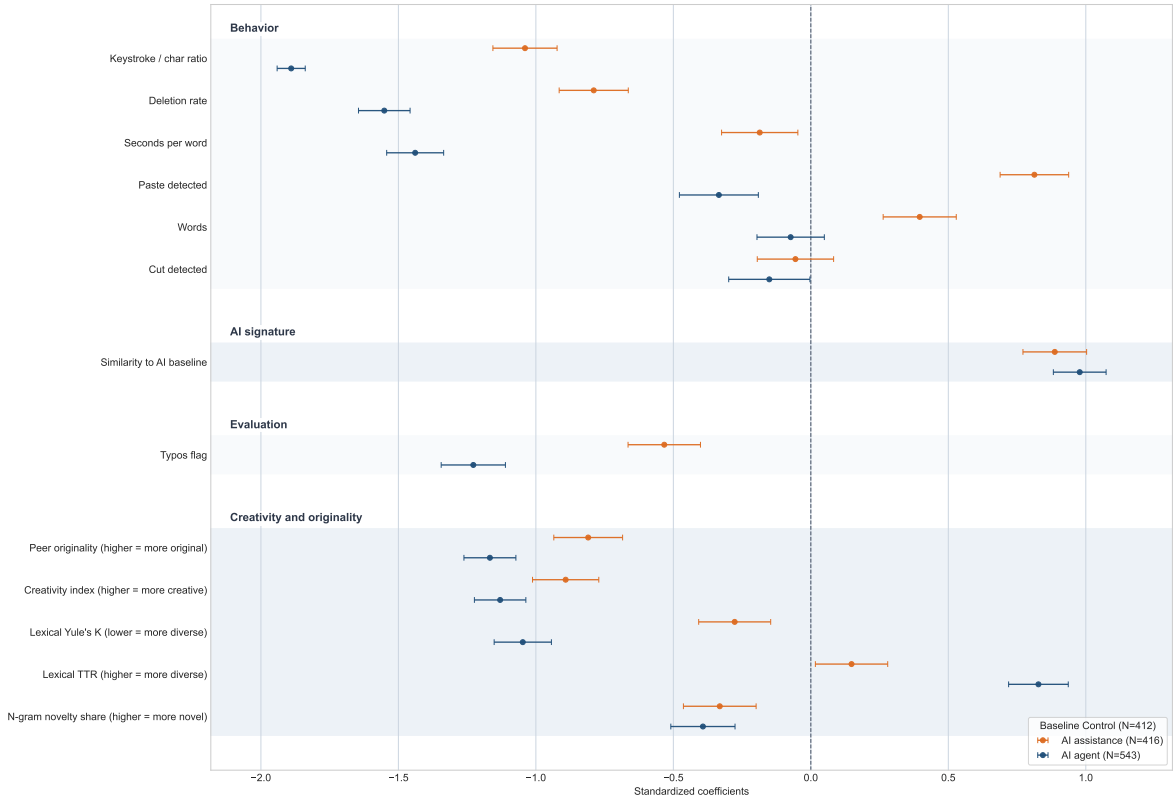

Figure 4: Standardized differences in summary-task outcomes for AI-assisted human responses and AI-agent responses, each relative to the human control group.
Note: The figure overlays two separate contrasts against the same baseline human control group. The *AI assistance* series compares human responses in the AI-only treatment to the control group. The *AI agent* series compares agent-generated responses to the same control group. Each point is estimated from a separate topic-adjusted OLS model with article-topic fixed effects, and horizontal bars indicate 95% HC1 confidence intervals. All dependent variables are standardized within the relevant comparison sample, so coefficients can be interpreted as differences in standard-deviation units. Positive coefficients indicate higher values than the human control group within that comparison, whereas negative coefficients indicate lower values. The outcome categories are the same as in Figure 2: *Behavior* captures writing effort, speed, and editing actions; *AI signature* captures similarity to an AI baseline summary; *Evaluation* includes the typo-flag indicator; and *Creativity and originality* captures originality, creativity, lexical diversity, and novelty.

the greatest benchmark similarity, and the lowest peer distinctiveness. At the same time, both the AI-assisted human group and the agent retain high scores in the evaluation (i.e, fewer typos). That combination is substantively important because it shows how AI-generated responses can look competent and well formed while still being produced under clearly different conditions.

Classifier Performance. Once AI-agent responses are added to the positive class, separation from baseline human control becomes substantially sharper. The best out-of-

sample models exceed 0.9 ROC-AUC, and, in this case, the best model is again a random forest, but in this case, it combines evaluation, AI signature (similarity with AI generic), and behavioral indicators. Our toolkit can again detect AI agent-generated answers with high accuracy and precision.

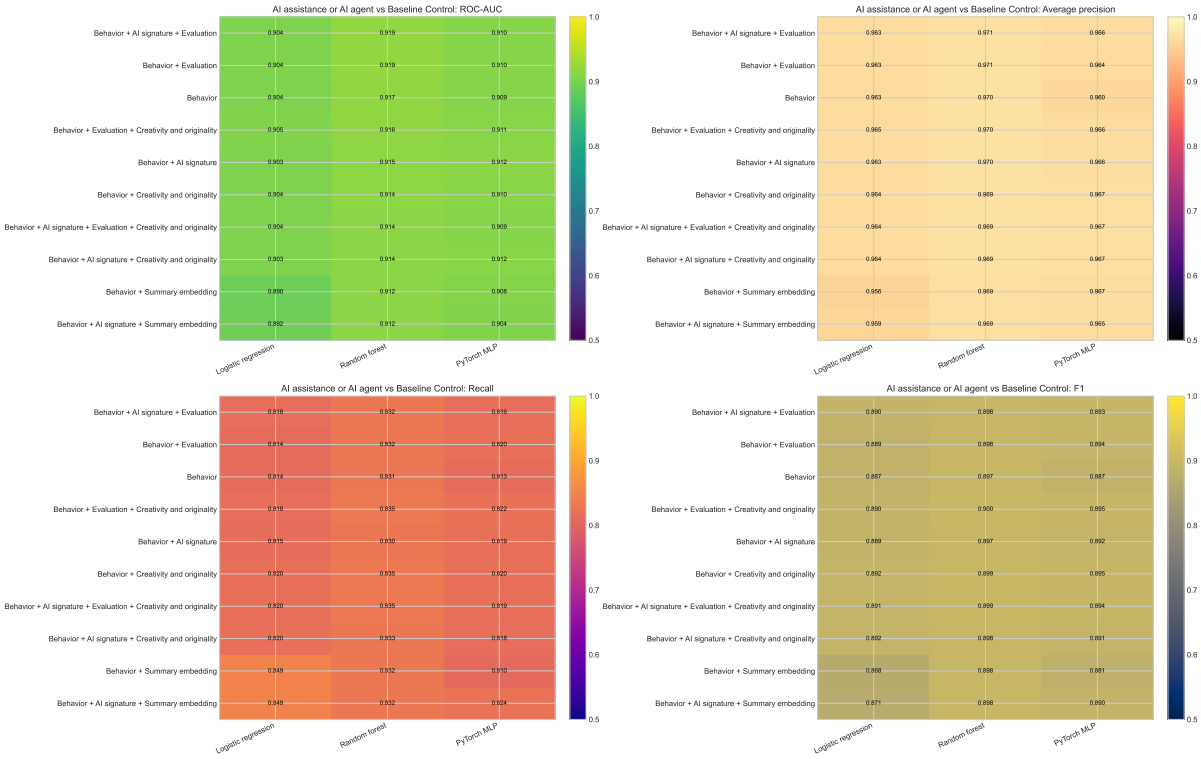

Figure 5: Top classifier combinations: AI assistance or AI agent vs. Baseline Control

Notes: The positive class combines human `AI assistance` responses and all AI agent responses; the comparison group is Baseline Control. The four panels report cross-validated ROC-AUC, average precision, recall, and F1. Rows show the top 10 feature-family combinations, ranked by classifier performance, and columns show the classifier used (logistic regression, random forest, and PyTorch MLP). Feature families are *Behavior*, *AI signature*, *Evaluation*, *Creativity and originality*, and *Summary embedding*. The summary embedding family is based on 384-dimensional summary-text embeddings from `sentence-transformers/all-MiniLM-L6-v2`. Larger values indicate better classifier performance. Sample sizes are $N = 412$ for Baseline Control and $N = 959$ for the combined AI group. The full heatmap with all feature-family combinations is reported in Appendix Figure 13.

7 Implications for Survey Research

The central implication of this note is that AI turns survey quality into an authentication problem. In an AI-rich environment, researchers can no longer treat a completed questionnaire, or even a polished open-ended response, as automatic evidence of human

comprehension, recall, judgment, or engagement. The contribution of this paper is not the claim that AI contamination exists, nor the simple observation that paradata can help. It is a survey-embedded authenticity workflow that uses open-ended items as observable evidence of human completion. The novelty lies in integrating front-end instrumentation, interpretable writing-process indicators, prompt-specific AI benchmarks, and model-based classification within an ordinary survey. Validated against both randomized AI-access conditions and a synthetic AI agent that completes the same instrument, the workflow shows how AI changes not only the final text but the response process itself.

That framing also clarifies the practical threat researchers now face. Some cases involve deception, in which fully automated agents impersonate respondents. Others involve delegation, in which real respondents outsource cognitively demanding parts of the survey to AI. Those problems overlap, but they do not have identical implications. Delegation is especially important for substantive inference because it can improve fluency, structure, and apparent engagement while reducing evidence of respondent effort. An answer can therefore look more articulate or more accurate while no longer measuring the respondent's own comprehension, recall, or reasoning. A longer or cleaner response is no longer enough. Researchers need evidence about how an answer was produced, not just what it says.

For survey practice, this implies a shift away from generic AI-text detection and toward study-specific authentication workflows. Researchers should collect process data such as paste attempts, keystrokes, deletions, pauses, and timing whenever ethically and technically feasible; generate benchmark AI answers for the exact prompt rather than rely on a detector divorced from the task; and evaluate how substantive conclusions change when likely AI-assisted cases are retained, down-weighted, or excluded. Our results also suggest that this is feasible in ordinary online surveys because behavior-centered models do much of the predictive work. The goal is not to impose a universal exclusion rule, but to build transparent, prompt-specific evidence about whether an open-ended response was drafted in the survey environment or outsourced to an external system.

The workflow should therefore be used diagnostically rather than only punitively. Flagged cases matter, but the flagging rate is itself information about the instrument. If a substantial share of respondents delegate open-ended tasks to AI, that may indicate more than opportunism; it may also reveal that the prompt is too burdensome, the incentive too weak, or the response format poorly matched to the kind of cognition the survey seeks to measure. The distinction between deception and delegation matters here. Fraudulent automation calls for stronger authentication and screening. Delegation often calls for redesign: shorter prompts, clearer instructions, lower-friction elicitation, or alternative response formats that make genuine participation easier than outsourcing. In that sense, the contribution of this paper is not only defensive. It also gives researchers a way to diagnose where survey design is inviting substitution.

These design choices must be paired with transparent reporting. Authors should state whether paste was permitted or restricted, what front-end instrumentation was active, how prompt-specific benchmark answers were generated, which indicators entered the workflow, whether evaluator features were collected directly or backfilled, and how any thresholds or models were selected. They should also archive the prompt templates and preprocessing code needed to reproduce the workflow. Without that documentation, readers cannot tell whether a flag is capturing AI-likely response production or simply unreported researcher discretion. Transparent disclosure is now part of the measurement argument itself, because claims about learning, understanding, persuasion, or justification depend on showing that the response process remained meaningfully human.

The broader lesson is that researchers must decide ex ante what they want open-ended items to measure. In some studies the target is unaided cognition. In others, assisted performance may itself be substantively meaningful. In still others, the relevant contrast is between the two. This paper's contribution is to make that choice operational by embedding authentication directly into the survey. When authenticity matters, even a small number of well-instrumented open-ended items can serve as diagnostic anchors for the broader questionnaire, revealing whether the survey is still capturing human effort or

has become vulnerable to outsourced production.

More broadly, open-ended responses should no longer be treated as isolated strings of text. They are response events produced within an interface, under incentives, and with observable traces. The task for survey methodology is therefore not simply to detect better instances of "AI text." It is to design surveys that can document whether a questionnaire was completed by a human, under what conditions, and with what level of human effort.

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A Appendix

A.1 Toolkit Pipeline

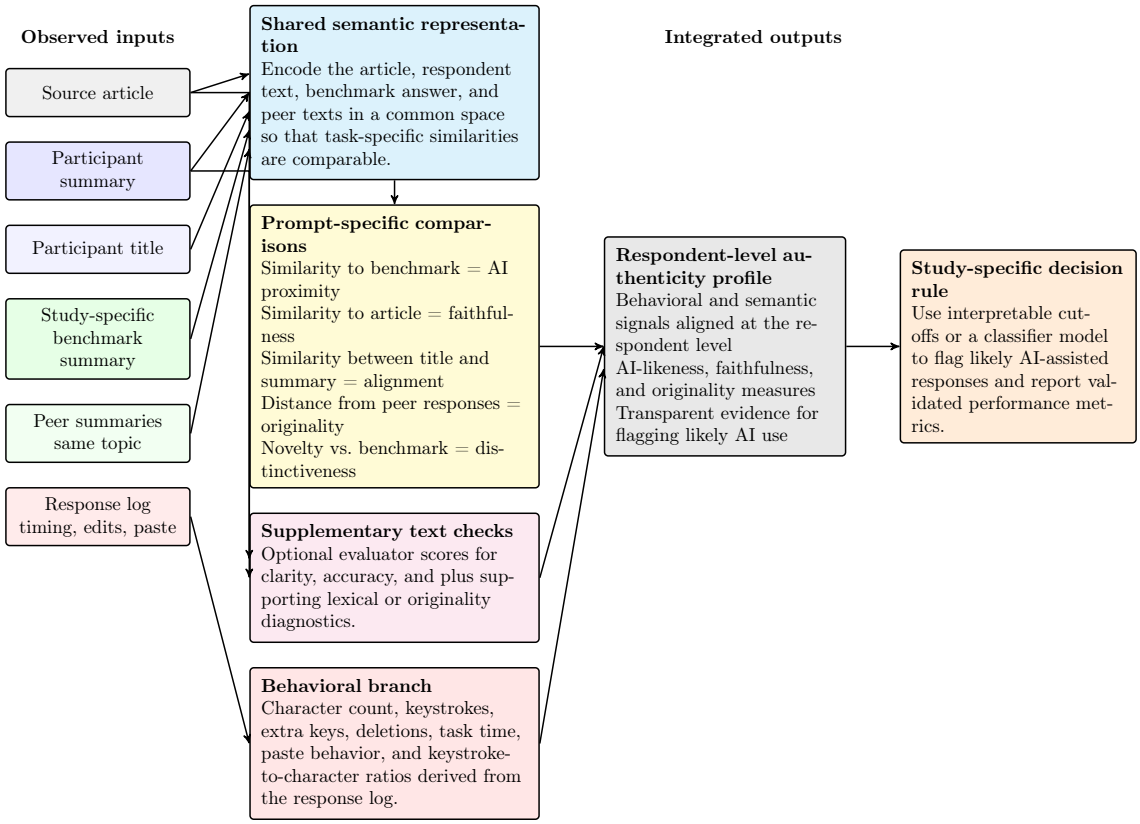

Figure 6: Workflow for the open-ended authentication toolkit. The pipeline shows how prompt-specific benchmarks, peer comparisons, and response-process data from instrumented open-ended questions are combined into a respondent-level authenticity profile that can be used to detect likely AI-assisted responding.

A.2 An Integrated Toolkit for AI-Response Detection

Our proposed toolkit addresses AI-assisted responding on three levels: (1) front-end deterrence, (2) real-time behavioral logging, and (3) post-hoc linguistic analysis. By combining these, we create a robust composite measure of "AI involvement" in a given open-ended response. Here we detail each component and the rationale behind it.

A.2.1 Front-End Manipulation: Copy-Paste Disablement

The first tool is a simple front-end intervention: disabling copy-paste functionality for certain survey questions. The logic is straightforward. A respondent leaning on an AI like ChatGPT typically would copy the AI's answer from another window and paste it into the survey text box. By technically blocking that action (via JavaScript embedded in the survey), we force respondents to manually type their answers in real time. This creates friction for those attempting to use external text generators. If a participant still

chooses to use AI, they must transcribe the AI-generated text keystroke by keystroke, a slower and more laborious process that our system can then detect through behavioral cues.

This manipulation serves two purposes: deterrence and detection. On one hand, it likely dissuades some users from cheating; if copy-paste is unavailable, the convenience of using AI drops, and more participants may answer in their own words. On the other hand, for those who persist in using AI (by re-typing an AI-provided answer), the act of doing so produces distinctive patterns we can capture.

In our implementation, if a paste is attempted, the script silently blocks it (nothing is inserted) and logs the event. The survey interface and question wording remain the same, preserving the user experience except for the disabled paste option. By being covert, this approach avoids alerting respondents to the specific detection mechanism (which could invite attempts to game it), while still upholding ethical standards – we do not prevent use of AI outright. Still, we ensure that if it is used, it becomes behaviorally evident.

A.2.2 Behavioral Monitoring: Keystroke Dynamics Logging

A cornerstone of our toolkit is the monitoring of typing behavior, or keystroke dynamics, during the answering process. We instrumented the survey text boxes with a logging script that records each key press, including timing, key codes, and whether it was an input or a deletion. These detailed paradata capture the process of composition for every open-ended response. From these logs, we derive features that can reveal whether an answer was composed through careful thought or generated elsewhere and transcribed.

Our approach builds on the idea that human writing is iterative, while AI-generated text (when transcribed by a human or pasted) appears in a more linear burst. Cognitive writing theory holds that humans plan, type, pause, revise, and edit in cycles; this manifests as starts and stops, deletions, and rephrasings in the keystroke record (Crossley et al. 2024). In contrast, if someone is copying an AI answer, the keystrokes tend to flow continuously as they transcribe or, in the case of a direct paste, a large chunk of text appears near-instantly with minimal backtracking.

Concretely, we log metrics such as: total keystrokes pressed, characters typed, backspace (delete) usage, pauses between keystrokes, and paste events. From these, we compute summary indicators. For example, a key metric is the ratio of characters to keystrokes (or its inverse). A genuinely composed answer typically involves many more keystrokes than the final character count (due to corrections and edits), yielding a low characters-per-keystroke ratio. An AI-generated answer that is pasted (or typed with almost no editing) will have a much higher ratio – possibly approaching 1.0 if the answer was typed out flawlessly or infinitely large if pasted in one go. We also look at "extra" keystrokes, defined as the total keystrokes minus the final number of characters in the submitted answer. This captures how much editing was done. A large number of extra keystrokes (relative to answer length) indicates the user spent effort revising the text (a human-like process), whereas near-zero extra keystrokes suggests the text arrived fully formed (suspicious of AI transcription).

Prior research strongly supports the efficacy of these indicators. Crossley et al. 2024 use keystroke logging to distinguish student essays written authentically versus copied from an outside source (e.g., an AI). They find that authentic writing showed frequent

pauses, insertions, and deletions, whereas transcribed text appeared in longer uninterrupted bursts. Their detection model achieved 99% accuracy in classifying an essay as original vs. copied based on features like pause frequency and revision counts. We leverage the same insight: how the text was generated is important information. Our toolkit flags responses with anomalous writing dynamics.

A.2.3 Back-End Linguistic Analysis: AI Similarity and Originality Checks

The third component of the toolkit analyzes the textual content of responses after submission, using modern NLP methods to evaluate their similarity to AI-generated text and their degree of originality. While behavioral data capture how a response was written, the linguistic layer provides independent evidence about what was written. By integrating both, we obtain a more reliable signal of whether an answer reflects genuine human reasoning or AI assistance.

LLM Similarity For each survey question, we generate reference answers using a large language model (GPT-3.5 or GPT-4) under the same prompt presented to participants. Each participant's text and the corresponding AI reference are embedded into a high-dimensional semantic space using the text-embedding-3-small model. The cosine similarity between these embeddings provides a continuous measure of semantic proximity to the AI baseline. Higher similarity indicates that the response resembles what an LLM would produce, whereas lower similarity reflects a more distinctive and idiosyncratic phrasing. We interpret this score as an indicator of "LLM-likeness." Although similarity alone cannot prove AI use, unusually high correspondence between a participant's summary and the model's output suggests reliance on or imitation of AI-generated content.

Originality and Faithfulness To complement this measure of LLM similarity, we compute several indicators of originality and topic adherence based on embeddings and lexical properties. Each response is compared to the article it summarizes, yielding a faithfulness score that reflects how closely the participant's writing captures the source material's main ideas. Simultaneously, the novelty of the response is measured as its cosine distance from the AI baseline embedding, indicating the extent to which the participant's phrasing diverges from model-generated language. A related alignment metric captures the coherence between each participant's title and summary by measuring their cosine similarity within the same embedding space.

We further examine textual originality through surface-level metrics. Lexical richness is assessed using the type–token ratio and Yule's K statistic, which together capture the diversity and repetition of vocabulary. We also compute the share of bigrams and trigrams in a response that do not appear in the corresponding article—an indicator of unique phrasing and paraphrasing effort (Kosmyna et al. 2025). To assess variation across participants exposed to the same article, we calculate peer originality, defined as the embedding distance between a participant's summary and the centroid of all summaries for that topic. Responses that are semantically distinct from the group mean, while remaining faithful to the article, are interpreted as more original.

Each of these features is standardized within topic to control for differences in article complexity and then combined into a composite Creativity Index. The index weights

novelty most heavily but also rewards faithfulness, internal coherence, and lexical variety. High values thus characterize responses that are both on-topic and written in the participant's own words, whereas low values indicate close resemblance to AI-generated or source text.

AI-as-Judge Evaluation In addition to quantitative similarity measures, we employ large language models as evaluators—or "AI judges"—to provide holistic assessments of the written responses. Each summary is presented to an LLM alongside the reference article, and the model produces ratings for clarity and coherence, factual accuracy, typo detection, and the degree to which the response appears to be written in the respondent's own words. This procedure mirrors recent advances in automated grading and text evaluation, where LLMs are used as consistent, low-variance evaluators across large datasets (Kosmyna et al. 2025). Although these judgments are not used as definitive classifiers of AI use, they serve as complementary indicators that capture dimensions of writing quality and originality that are difficult to express through embeddings or lexical statistics alone.

By combining these linguistic indicators with the behavioral data described earlier, the toolkit provides a multidimensional assessment of response authenticity. Rather than issuing a binary classification, it generates a detailed profile—capturing AI similarity, originality, and textual diversity—that allows researchers to identify suspicious cases and calibrate detection thresholds while minimizing false positives.

A.3 Data Collected

For each participant's summary response, we recorded the full text of the answer and a detailed keystroke log (including timestamps for each key event). Using the toolkit's back-end analysis, we generated a set of behavioral and linguistic indicators for every response. These measures jointly capture the process of composition (through keystroke dynamics) and the content of the written text (through embedding-based and lexical analysis). Specifically, we computed:

- **Character Count:** Total number of characters in the final submitted response.
- **Keystroke Count:** Total number of keys pressed, including letters, numbers, spaces, deletions, and other editing keys.
- **Extra Keystrokes:** Difference between keystroke count and character count, indicating the amount of revision and editing (extra keystrokes = keystrokes − characters).
- **Keystroke–Character Ratio:** Ratio of total key presses to final characters (keystrokes/characters), representing the extent of typing activity per retained character.
- **Paste Event:** Binary indicator denoting whether a paste action (`Ctrl/Command+V`) occurred during text entry.

- **Cosine Similarity to AI Answer:** Semantic similarity between the participant's response and a GPT-3.5–generated reference summary, computed using embeddings from the `text-embedding-3-small` model.
- **Cosine Similarity to Article:** Similarity between the participant's response and the original article text, reflecting topical faithfulness.
- **Novelty vs. AI:** Cosine distance ($1 -$ cosine similarity) from the AI baseline, measuring how distinct the participant's phrasing and ideas are from model-generated language.
- **Summary–Title Alignment:** Cosine similarity between a participant's summary and title embeddings, used as a measure of internal coherence.
- **Lexical Diversity:** Vocabulary richness estimated via the type–token ratio (TTR) and Yule's K statistic.
- **N-gram Novelty:** Proportion of bigrams and trigrams in the participant's response not present in the original article, capturing unique phrasing.
- **Peer Originality:** Embedding distance between a participant's summary and the centroid of all summaries written about the same article, indicating distinctiveness among peers.
- **AI-as-Judge Evaluation:** Structured ratings provided by a large language model (GPT-4) assessing clarity, coherence, accuracy, and use of one's own words when compared to the reference article.

All variables are standardized within topic to account for differences in article complexity, and they are combined in subsequent analyses to construct composite indicators such as the *Creativity Index* and overall measures of AI-likeness and response authenticity.

These measures were collected uniformly for all responses. The keystroke features and content features were then analyzed to see which best separate the two experimental groups (which serve as a proxy for likely AI use, since presumably AI-Allowed should have a higher incidence of AI-written answers). Because we experimentally manipulated one factor, we can interpret differences more cleanly than in an observational setting. In practice, a researcher using this toolkit in a real-world survey (without experimental conditions) would rely on a combination of metrics and perhaps a trained model to flag cases of AI-generated content.

A.4 A Synthetic AI Agent for Pre-Launch Survey Vulnerability Audits

An AI agent, in this setting, is a bounded software system that can perceive a survey page, represent the current task, generate answers conditional on a stable persona, and execute those answers in a browser without manual intervention. Rather than treating AI exposure as an abstract risk, researchers can use such an agent as a pre-launch stress test on their own instrument. If a controlled synthetic respondent can complete the survey

reliably, answer comprehension items from on-screen context, and produce plausible open-ended text at low cost, then the instrument is vulnerable to AI-mediated responding. The implementation described here was built for controlled benchmarking on an owned survey. Although the prototype targeted standard Qualtrics page structures, the logic is portable to other web surveys that expose ordinary HTML form elements.

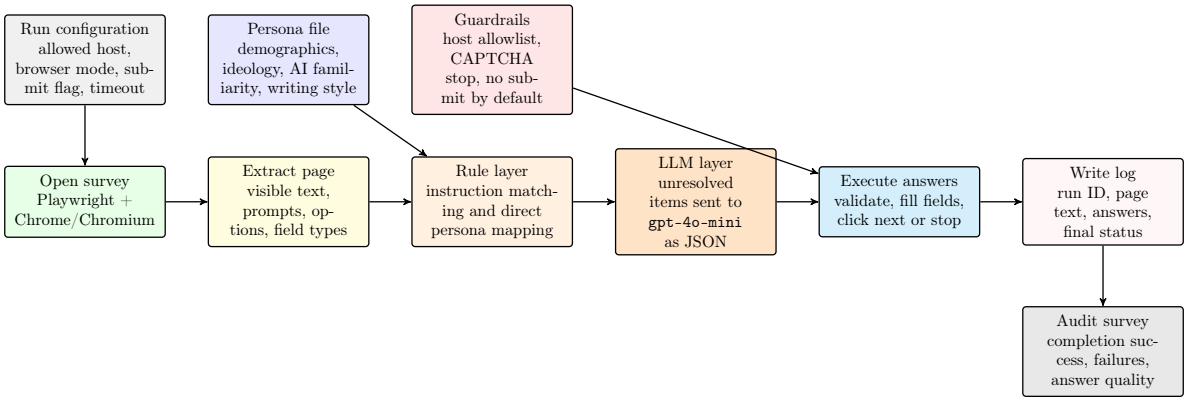

Figure 7: Synthetic-agent audit workflow. The agent opens the survey in a controlled browser, extracts visible questions into structured fields, answers simple items with rules, sends unresolved items to `gpt-4o-mini`, executes the validated answers in the browser, and stores a full run log for pre-launch auditing.

Using the agent as a template for vulnerability detection. Researchers can use this design as a pre-launch audit protocol. First, create a small pool of realistic personas rather than a single synthetic respondent. Second, run the instrument in dry mode and inspect the logs for unsupported widgets, broken parsers, or pages that the agent cannot interpret. Third, run a small submitted batch on the pilot version of the owned survey. Fourth, inspect whether the agent can pass attention checks, answer comprehension items from on-screen text, and generate plausible open-ended responses. Finally, compare completion success, answer diversity across personas, and the substantive quality of the generated text. A survey that a low-cost agent can complete reliably under these conditions should be treated as vulnerable to AI-mediated responding before field launch, even if ordinary fraud checks would not flag those cases.

Model and runtime. The benchmark agent was implemented in Python and used the OpenAI API together with Playwright browser automation. In the configuration used for the audit, the answer model was `gpt-4o-mini`, temperature was set to 0.5, the browser ran headless by default, the timeout was 15,000 milliseconds, and the agent retained up to four prior pages of textual context when answering later items. The core software stack was: `openai>=1.0.0` for model calls, `playwright==1.58.2` for browser control, Chrome/Chromium for page execution, JSON persona and configuration files for controlled variation, and JSON log files for post-run auditing. For larger batches, the same runner could be containerized and sharded across cloud jobs, with optional artifact upload through Google Cloud Storage.

What the agent actually received. The model did not receive unconstrained browsing instructions. Instead, it received a structured payload containing: (i) persona fields such as age group, gender, education, ethnicity, work status, occupation, ideology, AI familiarity, AI usage, description, and writing style; (ii) recent page text from earlier screens; (iii) current page text; (iv) prior answers; and (v) a list of unresolved fields with the prompt, field type, available options, and local question context. This design matters because it makes the agent auditable: researchers can see exactly what information the model had at each step.

Core instruction prompt. The system prompt supplied to the model was intentionally narrow and operational rather than conversational:

> "You are generating synthetic survey responses. Answer consistently as the provided persona. Follow the question instructions exactly. Return only valid JSON."

The response schema was similarly constrained:

```
{"answers": {"FIELD_ID": "string or list of strings"}}
```

Hybrid answer strategy. Not every answer was generated freely by the language model. Before calling the model, the agent first applied deterministic rules. If a question explicitly instructed the respondent to select a quoted option, the agent matched that option directly. If a field asked for demographics or background items represented in the persona file, the agent mapped the stored persona values onto the available response options. Only unresolved items were passed to the model. After the model responded, every returned answer was validated against the extracted response options before being entered into the survey. This hybrid design reduced unnecessary model calls and made the agent more consistent across repeated runs.

Survey parsing and supported question types. The page parser inspected visible question blocks and transformed them into a structured representation containing a field identifier, question identifier, prompt text, response type, required status, selector, response options, and local context. In the implementation used here, the agent supported five common classes of items: single-choice questions, multi-select questions, dropdown menus, text-entry boxes, and radio-matrix rows. Unsupported widgets were not silently guessed; instead, they appeared as failures or incomplete pages in the run log, which is itself useful information for a pre-launch vulnerability audit.

How the agent was run. A dry run opened the survey, answered pages, and stopped before the final submission button:

```
python -m survey_agent \
  --survey-url "YOUR_SURVEY_URL" \
  --persona-file configs/persona.json \
  --config-file configs/agent_config.json
```

Adding `-submit` allowed final submission on an owned survey. Adding `-count`, `-persona-pool-file`, and `-sleep-seconds` enabled small benchmark batches across multiple synthetic personas. A visible browser mode was also available through `-headful`, and a manual pause could be triggered at a CAPTCHA gate if the researcher wanted to inspect that failure mode.

Guardrails and audit value. Each run wrote a JSON log containing the final status, extracted page text, parsed fields, chosen answers, and error information when relevant. These safeguards are methodologically useful because the goal is not covert completion of arbitrary surveys, but controlled benchmarking of one's own instrument before deployment.

A.5 Additional Results

Figure 8 reports the extended benchmark coefficient plot for the original human sample, comparing each AI treatment arm separately to the same baseline control group. Positive coefficients indicate that the corresponding outcome is higher in the given AI arm than in the control group, while negative coefficients indicate lower values relative to control.

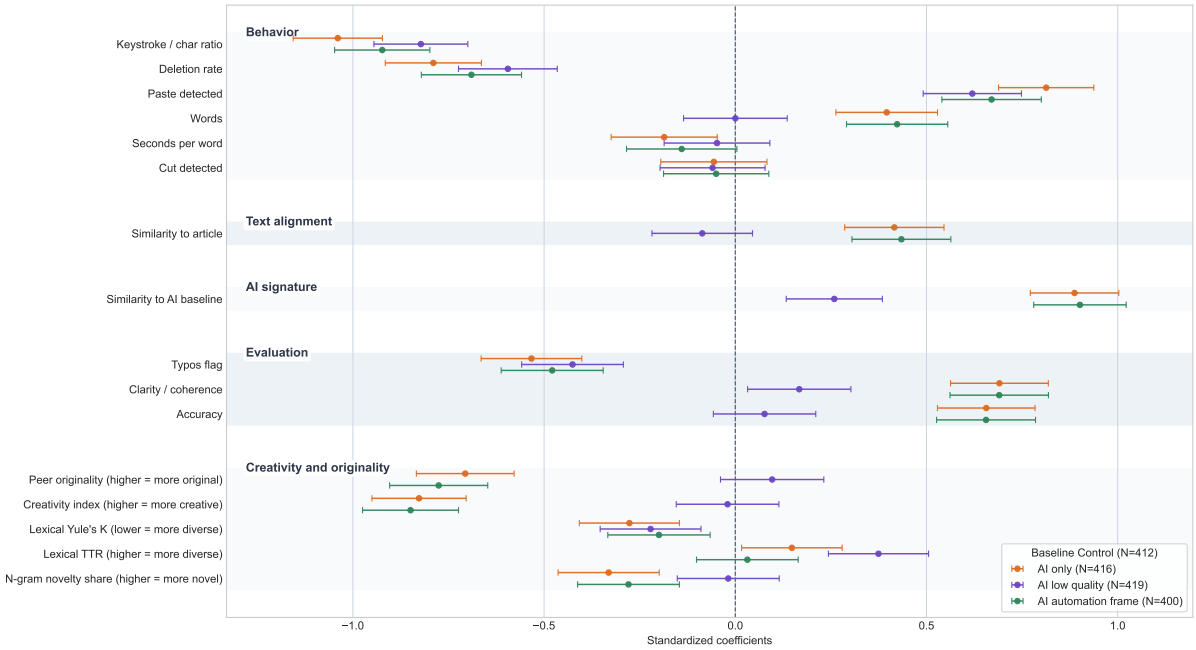

Figure 8: Extended benchmark coefficients for human AI-treatment arms relative to baseline control.

Notes: The figure reports standardized coefficients from separate topic-adjusted OLS regressions estimated on the original human dataset, with the same baseline control group used as the reference category in each comparison. The three series correspond to `AI_only` ($n = 416$), `AI_low` ($n = 419$), and `AI_long_term` ($n = 400$), each compared against baseline control ($n = 412$). Outcomes are grouped into the same extended categories as in the main coefficient plot: *Behavior*, *Text alignment*, *AI signature*, *Evaluation*, and *Creativity and originality*. The extended specification includes *Seconds per word* in Behavior, *Similarity to article* in Text alignment, and *Clarity / coherence* and *Accuracy* in Evaluation. Whiskers denote 95% confidence intervals. Positive values indicate higher levels of the measure relative to baseline control.

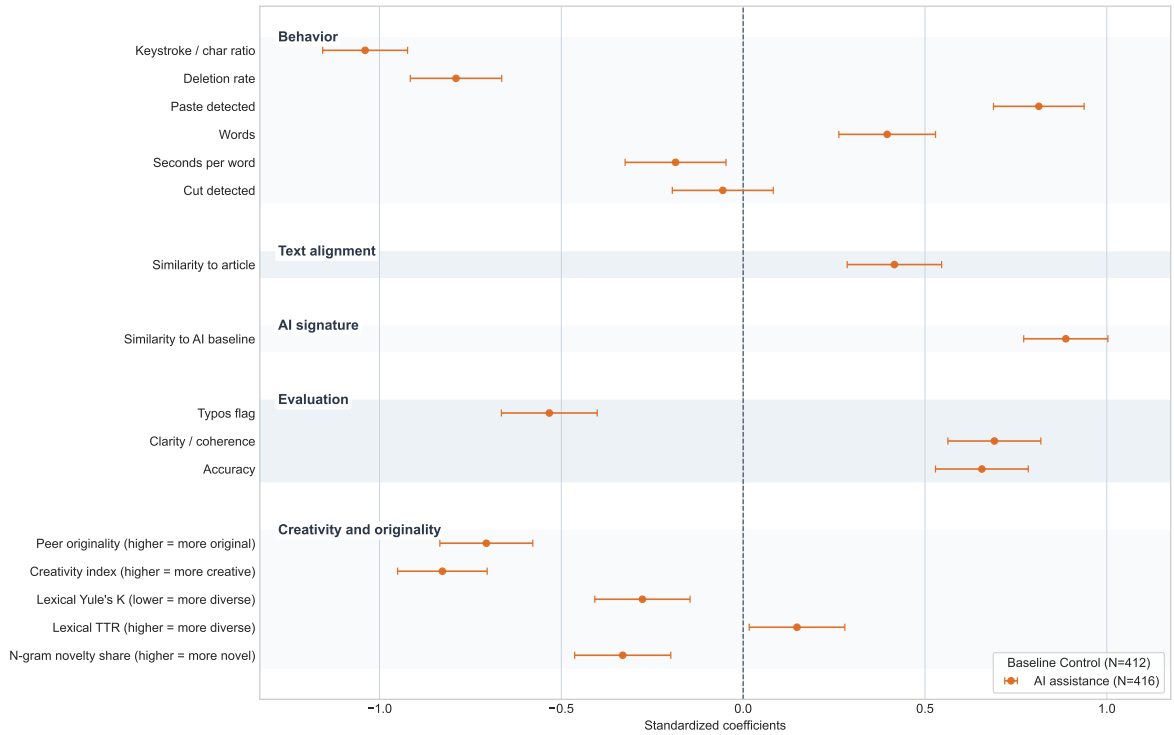

Figure 9: Extended standardized differences in summary-task outcomes between the human AI-only treatment and the human control group.
Note: This figure uses the same sample and specification as Figure 2: each coefficient comes from a separate topic-adjusted OLS regression with article-topic fixed effects, estimated on human `AI assisted` observations and control observations. Dependent variables are standardized, so coefficients are directly comparable across rows. Positive coefficients indicate that the AI-only group scores higher than the control group on the relevant outcome; negative coefficients indicate lower values. Relative to Figure 2, this extended specification adds a *Text alignment* category, where *Similarity to article* measures how closely the submitted summary tracks the source article, and expands the *Evaluation* category to include *Clarity / coherence* and *Accuracy*, where higher values indicate better evaluator assessments. As in Figure 2, the *Behavior* outcomes capture effort, speed, and copy/paste activity; the *AI signature* outcome captures similarity to an AI baseline; *Typos flag* is a binary indicator for whether typos were flagged; and the *Creativity and originality* outcomes capture originality, creativity, lexical diversity, and novelty.

A.6 Additional Metrics Classification

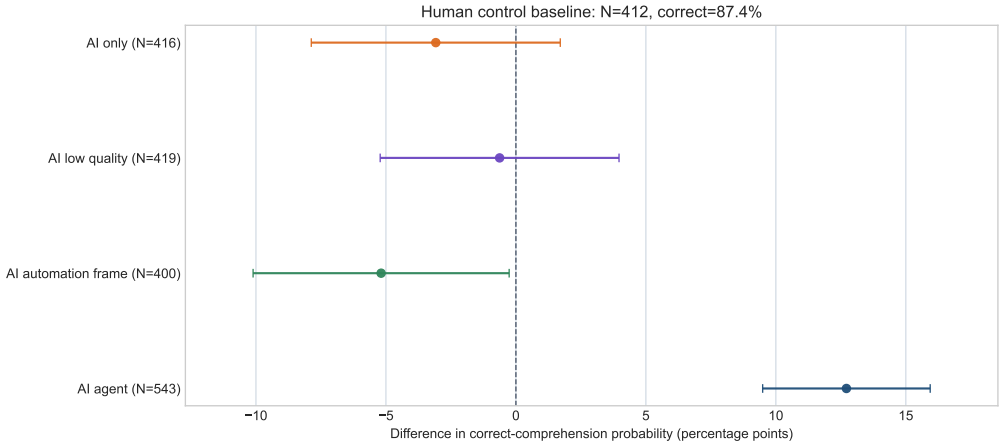

Figure 10: Comprehension accuracy across AI conditions relative to human control.
Notes: Each point compares one AI condition with the shared human control baseline in a separate linear probability model with article-topic fixed effects. Horizontal bars show 95% HC1 confidence intervals. Coefficients are reported in percentage points, so positive values indicate higher probability of answering the comprehension question correctly than in the human control group and negative values indicate lower probability. The four contrasts are `AI_only`, `AI_low`, `AI_long_term`, and the AI-agent benchmark. The human control baseline contains $N = 412$ observations and an 87.4% correct-comprehension rate.

Indicator	AI only				Any AI				AI assistance or Agent			
Indicator	ROC-AUC	Avg. prec.	Recall	F1	ROC-AUC	Avg. prec.	Recall	F1	ROC-AUC	Avg. prec.	Recall	F1
Behavior												
Seconds per word	0.601	0.642	0.668	0.591	0.586	0.831	0.669	0.712	0.836	0.939	0.779	0.816
Words	0.640	0.608	0.649	0.643	0.597	0.802	0.567	0.667	0.603	0.744	0.707	0.748
Deletion rate	0.762	0.806	0.697	0.681	0.730	0.898	0.683	0.754	0.883	0.944	0.853	0.874
Keystroke / char ratio	0.767	0.820	0.567	0.702	0.734	0.904	0.513	0.669	0.896	0.959	0.802	0.883
Paste detected	0.653	0.675	0.385	0.537	0.628	0.835	0.322	0.482	0.551	0.743	0.167	0.281
Cut detected	0.499	0.501	0.995	0.669	0.499	0.750	0.995	0.855	0.502	0.700	0.998	0.824
AI signature												
Similarity to AI baseline	0.724	0.793	0.565	0.646	0.668	0.879	0.520	0.648	0.737	0.905	0.643	0.759
Evaluation												
Typos flag	0.609	0.571	0.745	0.668	0.600	0.794	0.723	0.767	0.686	0.791	0.889	0.847
Creativity and originality												
Lexical TTR (higher = more diverse)	0.549	0.531	0.543	0.547	0.565	0.780	0.557	0.654	0.655	0.807	0.569	0.662
Lexical Yule's K (lower = more diverse)	0.637	0.628	0.668	0.633	0.610	0.821	0.640	0.714	0.740	0.867	0.712	0.768
N-gram novelty share (higher = more novel)	0.608	0.609	0.490	0.542	0.558	0.793	0.449	0.575	0.623	0.794	0.589	0.683
Peer originality (higher = more original)	0.699	0.778	0.567	0.622	0.633	0.865	0.521	0.639	0.747	0.906	0.644	0.750
Creativity index (higher = more creative)	0.723	0.795	0.575	0.642	0.656	0.875	0.517	0.643	0.748	0.907	0.642	0.758

Table 2: Standalone predictive power of indicators across AI-use classifiers

Notes: Each cell reports performance from a standalone cross-validated logistic-regression classifier estimated with a single indicator at a time. Columns report ROC-AUC, average precision, recall, and F1. *AI only* compares human `AI_only` responses to Baseline Control; *Any AI* compares all human AI-assistance arms (`AI_only`, `AI_low`, and `AI_long_term`) to Baseline Control; *AI assistance or Agent* combines human `AI_only` responses and all AI agent responses against Baseline Control. Indicators are grouped into *Behavior*, *AI signature*, *Evaluation*, and *Creativity and originality*, matching the summary-only benchmark coefficient plot. Baseline Control has $N = 412$; positive-class sample sizes are $N = 416$ for AI only, $N = 1235$ for Any AI, and $N = 959$ for AI assistance or Agent.

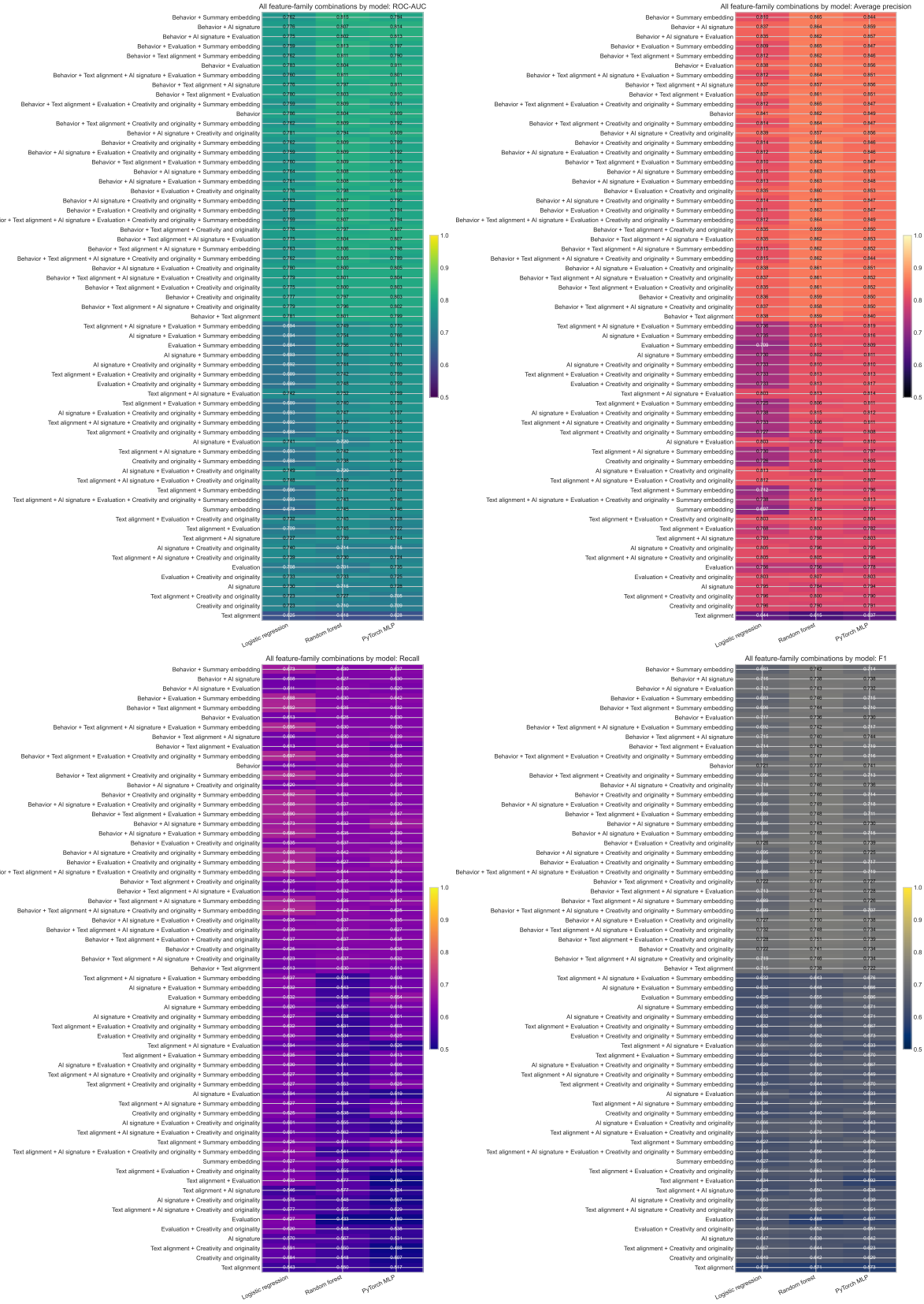

Figure 11: AI-only assistance versus baseline control: classifier performance across feature-family combinations.

Notes: The figure reports out-of-sample classifier performance for distinguishing human responses written with AI assistance (`AI_only`) from baseline human control responses. The analysis sample contains $N = 828$ observations ($n = 416$ AI-only, $n = 412$ control). Columns correspond to logistic regression, random forest, and PyTorch MLP models. Rows correspond to all possible combinations of the feature families *Behavior*, *Text alignment*, *AI signature*, *Evaluation*, *Creativity and originality*, and *Summary embedding*. The four panels report ROC-AUC, average precision, recall, and F1. Rows are ordered by the highest ROC-AUC achieved by any model for that feature-family combination. The *Summary embedding* family is based on a 384-dimensional sentence-transformer representation of the written summary (`all-MiniLM-L6-v2`). In this comparison, the strongest-performing bundle combines *Behavior* and

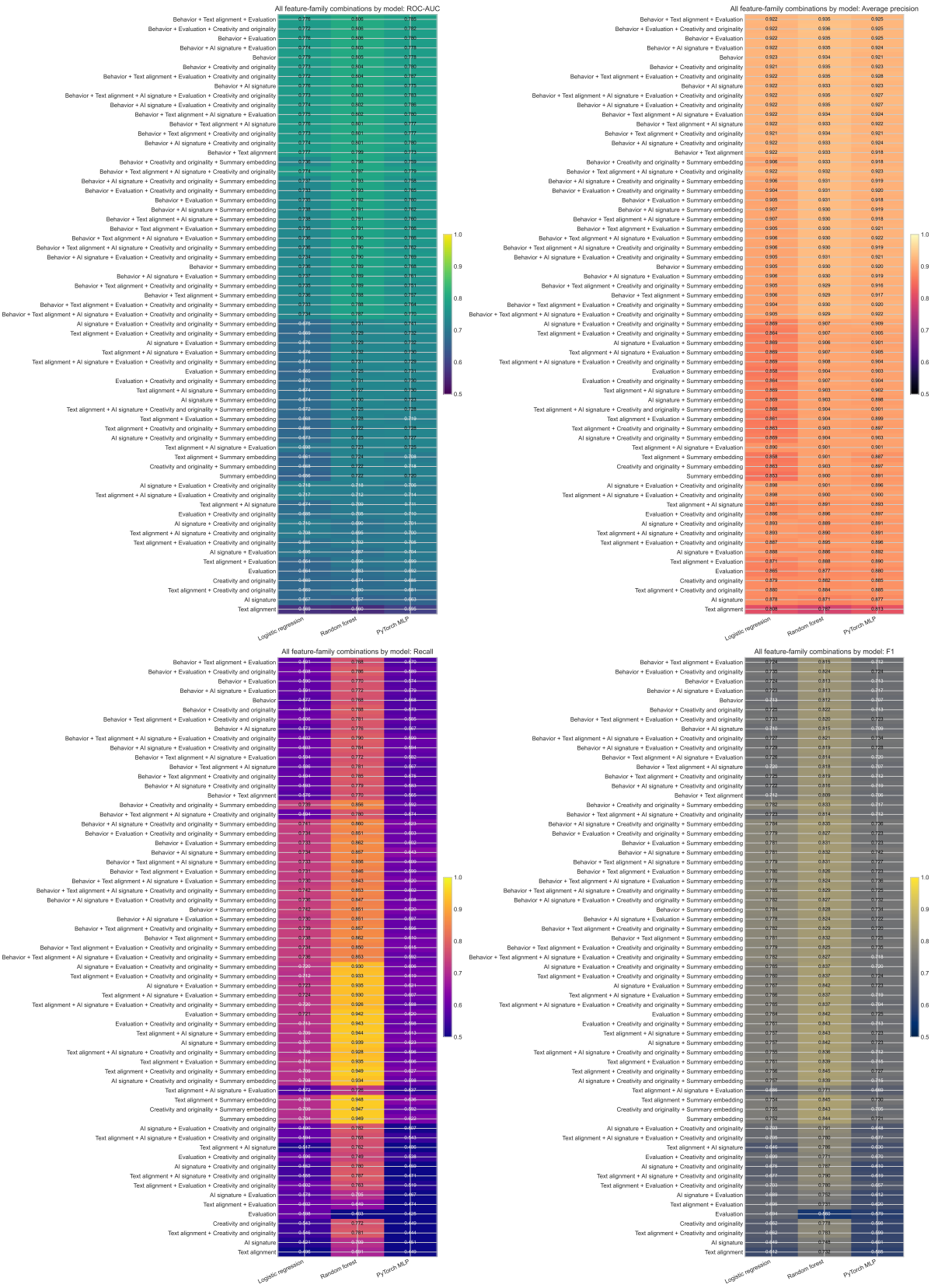

Figure 12: Any AI assistance versus baseline control: classifier performance across feature-family combinations.

Notes: The figure reports out-of-sample classifier performance for distinguishing any AI-assisted human response (including those receiving low quality AI) from baseline human control responses. The analysis sample contains $N = 1647$ observations ($n = 1235$ Any-AI, $n = 412$ control). Columns correspond to logistic regression, random forest, and PyTorch MLP models. Rows correspond to all possible combinations of the feature families *Behavior*, *Text alignment*, *AI signature*, *Evaluation*, *Creativity and originality*, and *Summary embedding*. The four panels report ROC-AUC, average precision, recall, and F1. Rows are ordered by the highest ROC-AUC achieved by any model for that feature-family combination. The *Summary embedding* family is based on a 384-dimensional sentence-transformer representation of the written summary (`all-MiniLM-L6-v2`). In this broader comparison, the best-performing bundle

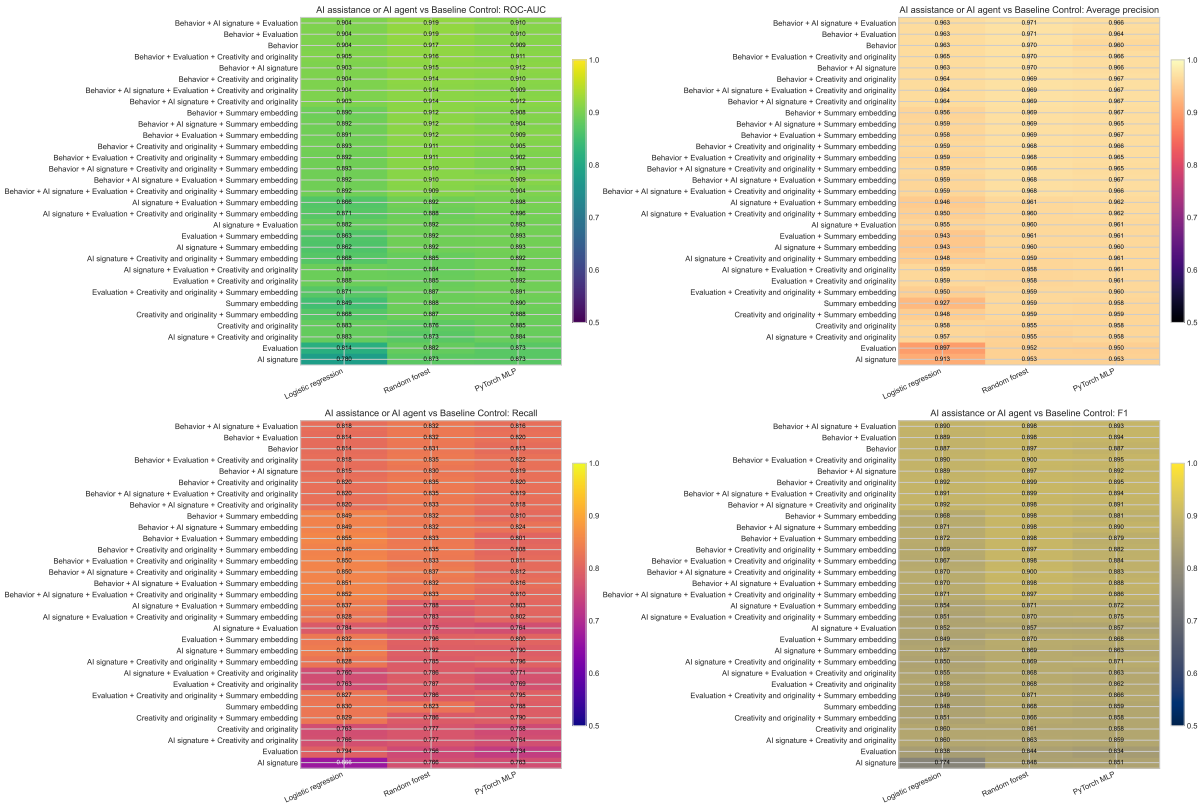

Figure 13: Classifier performance across feature-family combinations: AI assistance or AI agent vs. Baseline Control

Notes: The positive class combines human `AI assistance` responses and all AI agent responses; the comparison group is Baseline Control. The four panels report cross-validated ROC-AUC, average precision, recall, and F1. Rows show feature-family combinations, and columns show the classifier used (logistic regression, random forest, and PyTorch MLP). Feature families are *Behavior*, *AI signature*, *Evaluation*, *Creativity and originality*, and *Summary embedding*. The summary embedding family is based on 384-dimensional summary-text embeddings from `sentence-transformers/all-MiniLM-L6-v2`. Larger values indicate better classifier performance. Sample sizes are $N = 412$ for Baseline Control and $N = 959$ for the combined AI group.